



Università degli Studi di Padova

Department of Physics and Astonomy «Galileo Galilei»

Master degree in Physics

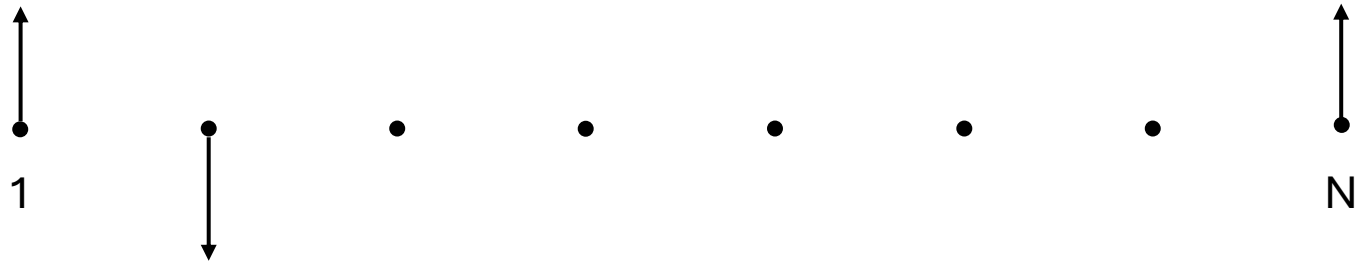
Mean Field Spin Glasses: the Parisi solution

Physics of Complex Systems

Padova, 29/01/2026

Alessandro Pennacchio

Sherrington-Kirkpatrick model



$\sigma = \{\sigma_i\}_{i=1,\dots,N} \qquad \sigma_i \in \{-1,1\}$

$$\mathcal{H}[\sigma] = -\underbrace{\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}}_{\sum_{i<j}} J_{ij}\,\sigma_i\,\sigma_j - h\sum_{i=1}^{N}\sigma_i \qquad J_{ij} \sim P(J) = \sqrt{\frac{N}{2\pi}}\; exp\left\{-\frac{N\,J^2}{2}\right\}$$

$$\mathcal{Z}_J = \sum_{\{\sigma\}} \exp\{-\beta\, \mathcal{H}[\sigma]\} \qquad \sum_{\{\sigma\}} = \prod_{i=1}^{N} \sum_{\sigma_i=-1,1}$$

$$f = \lim_{N\to\infty} f_J^{(N)} \qquad f_J^{(N)} = -\frac{1}{\beta}\frac{\ln \mathcal{Z}_J}{N}$$

$$\lim_{N\to\infty} \mathbb{P}\left[\left| f_J^{(N)} - \mathbb{E}\left[f_J^{(N)}\right]\right| \geq \theta\right] = 0 \qquad \forall\theta > 0 \qquad \text{(self averaging)}$$

$$\boxed{f = -\frac{1}{\beta} \lim_{N\to\infty} \frac{\mathbb{E}[\ln \mathcal{Z}_J]}{N}}$$

Replica trick

$$\mathcal{Z}_J^n = e^{n \ln \mathcal{Z}_J} \simeq 1 + n \ln \mathcal{Z}_J \qquad \Rightarrow \qquad \ln \mathcal{Z}_J = \lim_{n\to 0} \frac{\mathcal{Z}_J^n - 1}{n}$$

$$\Rightarrow \qquad \mathbb{E}[\ln \mathcal{Z}_J] = \lim_{n\to 0} \frac{\mathbb{E}[\mathcal{Z}_J^n] - 1}{n} = \lim_{n\to 0} \frac{\ln \mathbb{E}[\mathcal{Z}_J^n]}{n} , \qquad \ln(1+\delta) \simeq \delta , \qquad \delta = \mathbb{E}[\mathcal{Z}_J^n] - 1$$

$$f = -\frac{1}{\beta} \lim_{N\to\infty} \lim_{n\to 0} \frac{\ln \mathbb{E}[\mathcal{Z}_J^n]}{N\, n}$$

$$\boxed{f = -\frac{1}{\beta} \lim_{n\to 0} \lim_{N\to\infty} \frac{\ln \mathbb{E}[\mathcal{Z}_J^n]}{N\, n}}$$

$$\mathcal{Z}_J^n = \prod_{a=1}^{n} \mathcal{Z}_J = \left(\prod_{a=1}^{n} \prod_{i=1}^{N} \sum_{\sigma_i^a = -1,1} \right) \exp\left\{ \beta \sum_{a=1}^{n} \sum_{i<j} J_{ij}\, \sigma_i^a\, \sigma_j^a + \beta\, h \sum_{a=1}^{n} \sum_{i} \sigma_i^a \right\}$$

$$\mathbb{E}[\mathcal{Z}_J^n] = \int \left(\prod_{i<j} dJ_{ij}\, P(J_{ij}) \right) \left(\prod_{a} \prod_{i} \sum_{\sigma_i^a = -1,1} \right) \exp\left\{ \beta \sum_{a=1}^{n} \sum_{i<j} J_{ij}\, \sigma_i^a\, \sigma_j^a + \beta\, h \sum_{a=1}^{n} \sum_{i} \sigma_i^a \right\} =$$

$$= \left(\prod_{a} \prod_{i} \sum_{\sigma_i^a = -1,1} \right) \exp\left\{ \beta\, h \sum_{a} \sum_{i} \sigma_i^a \right\} \prod_{i<j} \underbrace{\int dJ_{ij} \exp\left\{ -\frac{N J_{ij}^2}{2} + \left(\beta \sum_{a} \sigma_i^a\, \sigma_j^a \right) J_{ij} \right\}}_{\text{Gaussian}}$$

$$\int dx \exp\left\{-\frac{A\,x^2}{2} + B\,x\right\} = \sqrt{\frac{2\pi}{A}} \exp\left\{\frac{B^2}{2\,A}\right\} \qquad A = N \qquad B = \beta \sum_a \sigma_i^a\,\sigma_j^a$$

$$\mathbb{E}[\mathcal{Z}_J^n] = \left(\prod_a \prod_i \sum_{\sigma_i^a=-1,1}\right) \exp\left\{\beta\, h \sum_a \sum_i \sigma_i^a\right\} \prod_{i<j} \exp\left\{\frac{\beta^2}{2\,N}\left(\sum_a \sigma_i^a\,\sigma_j^a\right)^2\right\}$$

$$= \left(\prod_a \prod_i \sum_{\sigma_i^a=-1,1}\right) \exp\left\{\beta\, h \sum_a \sum_i \sigma_i^a\right\} \exp\left\{\frac{\beta^2}{2\,N} \underbrace{\sum_{i<j} \sum_{a,b} \sigma_i^a\,\sigma_j^a\,\sigma_i^b\,\sigma_j^b}_{(*)}\right\}$$

$$\sum_{i<j} = \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} = \frac{1}{2} \sum_{i=1}^{N} \sum_{\substack{j=1 \\ J \neq i}}^{N} = \frac{1}{2} \sum_{i,j=1}^{N} (1 - \delta_{ij})$$

$$(*) = \frac{1}{2} \sum_{i,j=1}^{N} \sum_{a,b=1}^{n} \sigma_i^a \, \sigma_i^b \, \sigma_j^a \, \sigma_j^b - \frac{1}{2} \sum_{i=1}^{N} \sum_{a,b=1}^{n} \sigma_i^a \, \sigma_i^a \, \sigma_i^b \, \sigma_i^b$$

$$\sigma \in \{-1,1\} \qquad \Rightarrow \qquad \sigma^2 = 1$$

$$\Rightarrow \qquad (*) = \frac{1}{2} \sum_{a,b=1}^{n} \left(\sum_{i=1}^{N} \sigma_i^a \, \sigma_i^b \right)^2 - \frac{N \, n^2}{2}$$

$$\frac{1}{2}\sum_{i,j=1}^{N} = \sum_{i<j} + \frac{1}{2}\sum_{i,j=1}^{N} \delta_{ij}$$

$$(*) = \frac{1}{2}\sum_{a<b}\left(\sum_{i=1}^{N} \sigma_i^a\, \sigma_i^b\right)^2 + \sum_{a=1}^{n}\left(\sum_{i=1}^{N} \sigma_i^a\, \sigma_i^a\right)^2 - \frac{N\, n^2}{2} =$$

$$= \frac{1}{2}\sum_{a<b}\left(\sum_{i=1}^{N} \sigma_i^a\, \sigma_i^b\right)^2 + \frac{N^2\, n - N\, n^2}{2}$$

$$\mathbb{E}[\mathcal{Z}_J^n] = \left(\prod_a \prod_i \sum_{\sigma_i^a = -1,1} \right) \exp\left\{ \beta\, h \sum_a \sum_i \sigma_i^a \right\} \exp\left\{ \frac{\beta^2}{4} (N\, n - n^2) \right\} \prod_{a<b} \underbrace{\exp\left\{ \frac{\beta^2}{2\, N} \left(\sum_i \sigma_i^a\, \sigma_i^b \right)^2 \right\}}_{\exp\left\{ \frac{B^2}{2A} \right\}}$$

$$\int dx \exp\left\{ -\frac{A\, x^2}{2} + B\, x \right\} = \sqrt{\frac{2\pi}{A}} \exp\left\{ \frac{B^2}{2\, A} \right\}$$

$$A = N\, \beta^2$$

$$B \quad \rightarrow \quad B^{ab} = \beta^2 \sum_i \sigma_i^a\, \sigma_i^b$$

$$\mathbb{E}[\mathcal{Z}_J^n] = \left(\prod_a \prod_i \sum_{\sigma_i^a = -1,1} \right) \exp\left\{ \beta\, h \sum_a \sum_i \sigma_i^a \right\} \exp\left\{ \frac{\beta^2}{4} (N\, n - n^2) \right\} \times$$

$$\prod_{a<b} \left(\frac{N\, \beta^2}{2\pi} \right)^{\frac{1}{2}} \int dQ_{ab}\ \exp\left\{ -\frac{N\, \beta^2}{2} Q_{ab}^2 + \left(\beta^2 \sum_i \sigma_i^a\, \sigma_i^b \right) Q_{ab} \right\}$$

$$\boxed{\mathbb{E}[\mathcal{Z}_J^n] = \int \left(\prod_{a<b} dQ_{ab} \right) \exp\{-N\, S[Q,h]\} \qquad (N \to \infty)}$$

$$\boxed{S[Q,h] = -\frac{\beta^2\, n}{4} + \frac{\beta^2}{2} \sum_{a<b} Q_{ab}^2 - \ln \left[\left(\prod_a \sum_{\sigma^a = -1,1} \right) \exp\left\{ \beta\, h \sum_a \sigma^a + \beta^2 \sum_{a<b} \sigma^a\, \sigma^b\, Q_{ab} \right\} \right]}$$

Saddle point approximation

$$\mathbb{E}[\mathcal{Z}_J^n] = \exp\{-N\, S[Q^*, h]\} \qquad (N \to \infty)$$

$$Q^* \quad | \quad S[Q^*, h] = \max_Q S[Q, h] \qquad (n < 1)$$

$$\boxed{f = \frac{1}{\beta} \lim_{n \to 0} \frac{S[Q^*, h]}{n}}$$

Replica symmetric solution

$$S[Q_{ab}] = S[Q_{\pi(a)\pi(b)}]$$

Ansatz: $Q_{ab} = Q_{\pi(a)\pi(b)}$

$\Rightarrow$ $Q_{aa} = 0$ $Q_{ab} = q$ $a \neq b$

$$S[q,h] = -\frac{\beta^2 n}{4} + \underbrace{\frac{\beta^2}{2}\sum_{a<b} Q_{ab}^2}_{(1)} - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) \exp\left\{\beta\, h \sum_a \sigma^a + \underbrace{\beta^2 \sum_{a<b} \sigma^a\, \sigma^b\, Q_{ab}}_{(2)}\right\}\right]$$

$$(1) = q^2 \sum_{a<b} 1 = q^2 \frac{n(n-1)}{2} \simeq -q^2 \frac{n}{2} \qquad (n \to 0)$$

$$(2) = q \sum_{a<b} \sigma^a \sigma^b = q \left[\frac{1}{2} \left(\sum_a \sigma^a \right)^2 - \frac{1}{2} \sum_a \sigma^a \sigma^a \right] = q \left[\frac{1}{2} \left(\sum_a \sigma^a \right)^2 - \frac{n}{2} \right]$$

$$S[q,h] = -n \frac{\beta^2}{4} (1-q)^2 - \ln \left[\left(\prod_a \sum_{\sigma^a = -1,1} \right) \exp\left\{ \beta\, h \sum_a \sigma^a \right\} \underbrace{\exp\left\{ \frac{\beta^2 q}{2} \left(\sum_a \sigma^a \right)^2 \right\}}_{\exp\left\{ \frac{B^2}{2A} \right\}} \right]$$

$$A = 1 \qquad B = \beta \sqrt{q} \sum_a \sigma^a$$

$$S[q,h] = -n\frac{\beta^2}{4}(1-q)^2 - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right)\exp\left\{\beta\, h \sum_a \sigma^a\right\}\int \frac{dz}{\sqrt{2\pi}}\exp\left\{-\frac{z^2}{2} + \left(\beta\sqrt{q}\sum_a \sigma^a\right)z\right\}\right] =$$

$$= -n\frac{\beta^2}{4}(1-q)^2 - \ln\left[\int \frac{dz}{\sqrt{2\pi}}\exp\left\{-\frac{z^2}{2}\right\}\prod_a \underbrace{\sum_{\sigma^a=-1,1}\exp\{(\beta\, h + \beta\sqrt{q}\, z)\,\sigma^a\}}_{2\cosh(\beta\, h + \beta\sqrt{q}\, z)}\right]$$

$$\boxed{S[q,h] = -n\left\{\frac{\beta^2}{4}(1-q)^2 + \int \frac{dz}{\sqrt{2\pi}}\exp\left\{-\frac{z^2}{2}\right\}\ln[2\cosh(\beta\, h + \beta\sqrt{q}\, z)]\right\} \qquad (n \to 0)}$$

$$\tilde{f}(q,h) = \frac{1}{\beta} \lim_{n\to 0} \frac{S[q,h]}{n} = -\left\{ \frac{\beta}{4}(1-q)^2 + \frac{1}{\beta} \int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \ln[2\cosh(\beta\, h + \beta\, \sqrt{q}\, z)] \right\}$$

$$\frac{\partial \tilde{f}(q,h)}{\partial q} = \frac{\beta}{2}(1-q) - \frac{1}{2\sqrt{q}} \int \frac{dz}{\sqrt{2\pi}} \underbrace{z \exp\left\{-\frac{z^2}{2}\right\}}_{-\frac{d}{dz}\exp\left\{-\frac{z^2}{2}\right\}} \mathrm{tgh}(\beta\, h + \beta\, \sqrt{q}\, z) =$$

$$= -\frac{\beta}{2}\left\{ q - \int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \mathrm{tgh}^2(\beta\, h + \beta\, \sqrt{q}\, z) \right\}$$

$$\left[\frac{\partial \tilde{f}(q,h)}{\partial q}\right]_{q=q*} = 0$$

$$\boxed{q^* = \int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \mathrm{tgh}^2\left(\beta\, h + \beta\, \sqrt{q^*}\, z\right)}$$

$$\boxed{f = \tilde{f}(q^*, h) = -\left\{\frac{\beta}{4}(1-q^*)^2 + \frac{1}{\beta}\int \frac{dz}{\sqrt{2\pi}}\, \exp\left\{-\frac{z^2}{2}\right\} \ln[2\cosh\left(\beta\, h + \beta\, \sqrt{q^*}\, z\right)]\right\}}$$

$$h = 0 \qquad q^* = \int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \mathrm{tgh}^2\left(\beta \sqrt{q^*}\, z\right)$$

$q^* = 0$ is a solution

$$T \to +\infty \qquad (\beta \to 0^+) \qquad q^* = 0$$

$$\tilde{f}(q,0) = -\left\{\frac{\beta}{4}(1-q)^2 + \frac{1}{\beta}\int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \ln[2\cosh(\beta\sqrt{q}\, z)]\right\} = -\left(\frac{\beta}{4} + \frac{\ln 2}{\beta}\right) - \frac{\beta}{4}(1-\beta^2)\, q^2 + \mathcal{O}(q^3)$$

$$\beta_c = 1 \qquad \begin{cases} \beta < \beta_c & \Rightarrow \quad q^* = 0 \\ \beta > \beta_c & \Rightarrow \quad q^* \neq 0 \quad \text{(spin glass phase)} \end{cases}$$

$$T \to 0^+ \qquad (\beta \to +\infty)$$

$$q^* = 1 - \int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \operatorname{sech}^2\left(\beta \sqrt{q^*}\, z\right)$$

$$y = \beta \sqrt{q^*}\, z \qquad z = \frac{y}{\beta \sqrt{q^*}} \to 0 \qquad (\beta \to \infty)$$

$$q^* \simeq 1 - \int \frac{dy}{\sqrt{2\pi}\, \beta \sqrt{q^*}} \operatorname{sech}^2(y) = 1 - \frac{2}{\sqrt{2\pi}\, \beta \sqrt{q^*}}$$

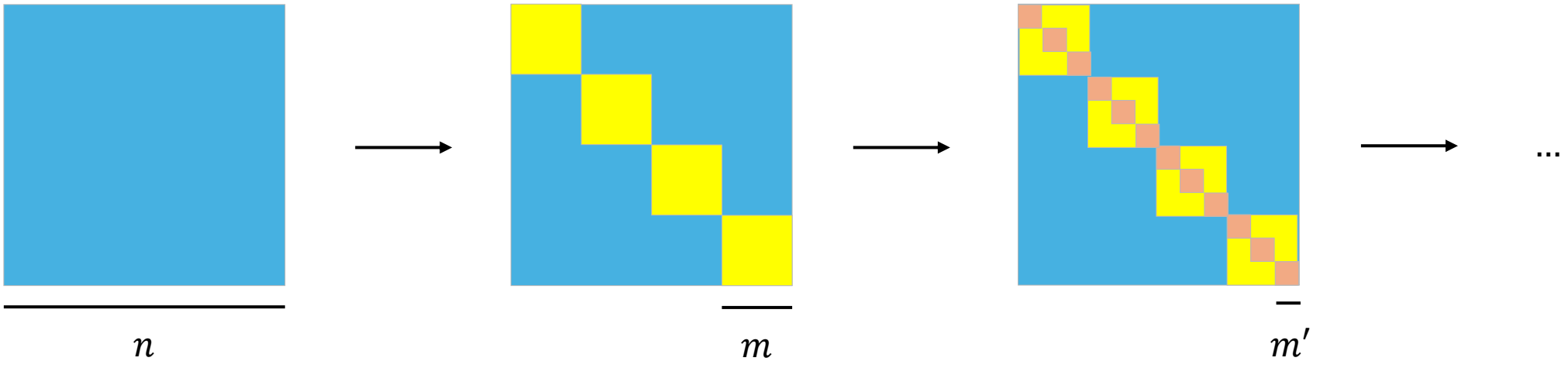
$$q^* \simeq 1 - \sqrt{\frac{2}{\pi}} \frac{1}{\beta}$$

$$f = -\left\{\frac{\beta}{4}(1-q^*)^2 + \frac{1}{\beta}\int \frac{dz}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \ln\left[2\cosh\left(\beta\sqrt{q^*}\, z\right)\right]\right\} \simeq -\frac{\beta}{4}(1-q^*)^2$$

$$s = -\frac{\partial f}{\partial T} = \beta^2 \frac{\partial f}{\partial \beta} \simeq -\frac{\beta^2}{4}(1-q^*)^2$$

$$\boxed{s(0) = -\frac{1}{2\pi} \approx -0.159 < 0}$$

Breaking replica symmetry

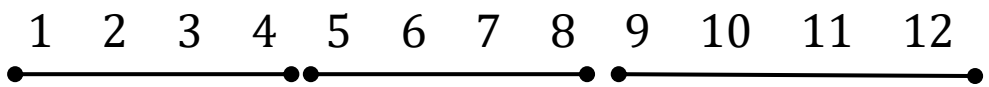


1-RSB

$$Q_{ab} = \begin{cases} q_1 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right) \\ q_0 & \text{if} \quad I\left(\frac{a}{m}\right) \neq I\left(\frac{b}{m}\right) \end{cases} \qquad 1 \leq m \leq n \xrightarrow{n \to 0} m \in [0,1]$$

E.g. $I(1.73) = 2 = I(2)$

E.g. $n = 12\,, \quad m = 4$



$$I\left(\frac{3}{4}\right) = 1 = I\left(\frac{4}{4}\right) \qquad I\left(\frac{11}{4}\right) = 3$$

$$S[q,h] = -\frac{\beta^2\, n}{4} + \underbrace{\frac{\beta^2}{2}\sum_{a<b} Q_{ab}^2}_{(1)} - \ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) \exp\left\{\beta\, h \sum_a \sigma^a + \underbrace{\beta^2 \sum_{a<b} \sigma^a\, \sigma^b\, Q_{ab}}_{(2)}\right\}\right]$$

$$\sum_{a<b} = \frac{1}{2}\left[\sum_{\substack{a,b=1 \\ I\left(\frac{a}{m}\right)=I\left(\frac{b}{m}\right)}}^{n} + \sum_{\substack{a,b=1 \\ I\left(\frac{a}{m}\right)\neq I\left(\frac{b}{m}\right)}}^{n} - \sum_{a,b=1}^{n} \delta_{ab}\right]$$

$$(1) = \frac{1}{2}\left[q_1^2\, n\, m + q_0^2\, n\, (n-m) - q_1^2\, n\right] \simeq \frac{n}{2}\left[-m\, q_0^2 - (1-m)\, q_1^2\right] \qquad (n \to 0)$$

$$(2) = \frac{1}{2}\left[q_1 \sum_{\substack{a,b=1 \\ I\left(\frac{a}{m}\right)=I\left(\frac{b}{m}\right)}}^{n} \sigma^a \sigma^b + q_0 \sum_{\substack{a,b=1 \\ I\left(\frac{a}{m}\right)\neq I\left(\frac{b}{m}\right)}}^{n} \sigma^a \sigma^b - q_1 \sum_{a=1}^{n} \sigma^a \sigma^a \right] =$$

$$= \frac{1}{2}\left[q_0 \sum_{a,b=1}^{n} \sigma^a \sigma^b + (q_1 - q_0) \underbrace{\sum_{\substack{a,b=1 \\ I\left(\frac{a}{m}\right)=I\left(\frac{b}{m}\right)}}^{n} \sigma^a \sigma^b}_{(*)} + -q_1 n \right]$$

$$(*) = \sum_{k=1}^{n/m} \sum_{a,b=1}^{m} \sigma^{a+m(k-1)}\, \sigma^{b+m(k-1)} = \sum_{k=1}^{n/m} \left(\sum_{a=1}^{m} \sigma^{a,k} \right)^2 \qquad \sigma^{a,k} = \sigma^{a+m(k-1)}$$

E.g. 1 2 3 4 5 6 7 8 9 10 11 12

$$11 = 3 + 4 \times (3 - 1)$$
$$\uparrow \quad \uparrow \quad \uparrow$$
$$a \quad m \quad k-1$$

$$(2) = \frac{1}{2} \left[q_0 \left(\sum_{a=1}^{n} \sigma^a \right)^2 + (q_1 - q_0) \sum_{k=1}^{n/m} \left(\sum_{a=1}^{m} \sigma^{a,k} \right)^2 - q_1\, n \right]$$

$$S[q,h] = -\frac{\beta^2 n}{4}[1 + m\, q_0^2 + (1-m)q_1^2 - 2\, q_1] -$$

$$-\ln\left[\left(\prod_a \sum_{\sigma^a=-1,1}\right) \underbrace{\exp\left\{\frac{\beta^2}{2}\, q_0 \left(\sum_{a=1}^{n} \sigma^a\right)^2\right\}}_{(1)} \prod_{k=1}^{n/m} \underbrace{\exp\left\{\frac{\beta^2}{2}\,(q_1 - q_0)\left(\sum_{a=1}^{m} \sigma^{a,k}\right)^2\right\}}_{(2)}\right]$$

$$(1) = \exp\left\{\frac{B^2}{2A}\right\} \qquad A = 1 \qquad B = \beta\sqrt{q}\sum_a \sigma^a$$

$$(2) = \exp\left\{\frac{B^2}{2A}\right\} \qquad A = \frac{1}{q_1 - q_0} \qquad B^k = \beta \sum_{a=1}^{m} \sigma^{a,k}$$

$$S[q,h] \simeq -\frac{\beta^2 n}{4}[1 + m\, q_0^2 + (1-m)q_1^2 - 2\, q_1] -$$

$$-\frac{n}{m}\int \frac{dz}{\sqrt{2\pi\, q_0}} \exp\left\{-\frac{z^2}{2\, q_0}\right\} \ln\left[\int \frac{dy}{\sqrt{2\pi\,(q_1 - q_0)}} \exp\left\{-\frac{y^2}{2(q_1 - q_0)}\right\} [2\cosh(\beta(z+y))]^m\right]$$

$$\tilde{f}(q_1, q_0, m) = \frac{1}{\beta} \lim_{n\to 0} \frac{S[q_1, q_0, m]}{n}$$

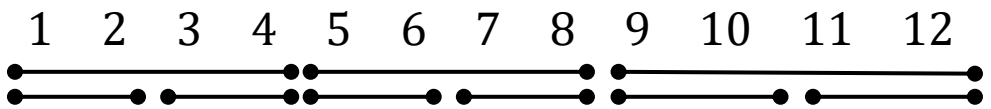
$$f = \max_{q_1, q_0, m} \tilde{f}(q_1, q_0, m\,)$$

$$\boxed{0 > s_{1-RSB}(0) \approx -0.01 > s_{RS}(0) \approx -0.159}$$

2-RSB

$$Q_{ab} = \begin{cases} q_2 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right), \quad I\left(\frac{a}{m'}\right) = I\left(\frac{b}{m'}\right) \\ q_1 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right), \quad I\left(\frac{a}{m'}\right) \neq I\left(\frac{b}{m'}\right) \\ q_0 & \text{otherwise} \end{cases} \qquad 1 \leq m' \leq m \leq n$$

E.g.: $n = 12$, $m = 4$, $m' = 2$



$$I\left(\frac{5}{4}\right) = 2 = I\left(\frac{7}{4}\right)$$

$$I\left(\frac{5}{2}\right) = 3 \neq 4 = I\left(\frac{7}{2}\right)$$

K-RSB

$$Q_{ab} = \begin{cases} q_k & \text{if} \quad I\left(\frac{a}{m_i}\right) = I\left(\frac{a}{m_i}\right) \quad i = 1, \dots, k \\ q_{k-1} & \text{else if} \quad I\left(\frac{a}{m_i}\right) = I\left(\frac{a}{m_i}\right) \quad i = 1, \dots, k-1 \\ \vdots & \\ q_1 & \text{else if} \quad I\left(\frac{a}{m_1}\right) = I\left(\frac{a}{m_1}\right) \\ q_0 & \text{otherwise} \end{cases}$$

$$1 \leq m_k \leq m_{k-1} \leq \cdots \leq m_1 \leq n$$

$$\rightarrow \quad S[q_k, \dots, q_0, m_k, \dots, m_1], \qquad \tilde{f}(q_k, \dots, q_0, m_k, \dots, m_1)$$

$$\rightarrow \quad f = \max_{q_k, \dots, q_0, m_k, \dots, m_1} \tilde{f}(q_k, \dots, q_0, m_k, \dots, m_0)$$

$$\boxed{0 > s_{K-RSB}(0) > s_{(K-1)-RSB}(0) > \cdots > s_{1-RSB}(0) > s_{\mathrm{RS}}(0)}$$

Full-RSB

$Q_{ab} = q_k$ if $I\left(\frac{a}{m_i}\right) = I\left(\frac{a}{m_i}\right)$ $i = 1, \dots, k$ $1 \leq m_k \leq m_{k-1} \leq \cdots \leq m_1 \leq n$

$n \to 0$ $\Rightarrow$ $m_0 \leq m_1 \leq \cdots \leq m_{k-1} \leq m_k \leq m_{k+1}$ $(m_0 = n\,,$ $m_{k+1} = 1)$

$q(x) = q_i$ $m_i < x < m_{i+1}$ $i = 0, \dots, k$

$n \to 0$ $\Rightarrow$ $x \in [0,1]$

$K \to \infty$

$\to$ $\tilde{f}[q(x)]$

$\to$ $f = \max \tilde{f}[q(x)]$

$$f = -\frac{\beta}{4}\left[1 + \int_0^1 dx\, q^2(x) - 2\, q(1)\right] - \frac{1}{\beta}\int \frac{du}{\sqrt{2\pi}} \exp\left\{-\frac{u^2}{2}\right\} f_0\left(0, \sqrt{q(0)}\, u\right)$$

$$\frac{\partial f_0(x,h)}{\partial x} = -\frac{1}{2}\frac{dq(x)}{dx}\left[\frac{\partial^2 f_0(x,h)}{\partial h^2} + x\left(\frac{\partial f_0(x,h)}{\partial h}\right)^2\right] \qquad f_0(1,h) = \ln[2\cosh(h)]$$

$$s_{full-RSB}(0) = 0$$

Pure states

$$f = \frac{1}{\beta} \lim_{n \to 0} \frac{S[Q^*, h]}{n} \qquad Q^* \quad | \quad S[Q^*, h] = \max_Q S[Q, h] \qquad S[Q_{ab}] = S[Q_{\pi(a)\pi(b)}]$$

→ pure states (labelled by α)

$$\langle \dots \rangle = \sum_\alpha \langle \dots \rangle_\alpha$$

$$q_{\alpha\beta} = \frac{1}{N}\sum_{i=1}^{N} \langle \sigma_i \rangle_\alpha \langle \sigma_i \rangle_\beta$$

$$P_J(q) = \sum_{\alpha,\beta} w_\alpha w_\beta \ \delta(q_{\alpha\beta} - q)$$

$$P(q) = \mathbb{E}\left[P_J(q)\right] = \lim_{n\to 0} \frac{1}{n(n-1)} \sum_{\substack{a,b=1 \\ a\neq b}}^{n} \delta(Q_{ab} - q) =$$

$$= \int_0^1 dx\, \delta(q(x) - q)$$

$$\boxed{x(q) = \int_0^q dq' P(q')}$$

Fluctuations of $P_J(q)$

$$\boxed{\mathbb{E}\left[P_J(q_1)\,P_J(q_2)\right] = \frac{1}{3}P(q_1)\delta(q_1 - q_2) + \frac{2}{3}P(q_1)P(q_2)}$$

Stochastic stability

Stochastic stability principle: $\rho(F)$ stable under indipendent random increments

1-RSB

$$Q_{ab} = \begin{cases} q_1 & \text{if} \quad I\left(\frac{a}{m}\right) = I\left(\frac{b}{m}\right) \\ q_0 & \text{if} \quad I\left(\frac{a}{m}\right) \neq I\left(\frac{b}{m}\right) \end{cases} \qquad 1 \leq m \leq n \xrightarrow{n \to 0} m \in [0,1]$$

$$\boxed{\rho(F) \propto e^{\beta\, m\, F}}$$

$$\sum_{\alpha} \rho(F_\alpha) = 1 \qquad \Rightarrow \qquad m < 1$$

Ultrametricity

Topology of the space of pure states

$$P(q_1, q_2, q_3) = \mathbb{P}\left[q_{\alpha_1\alpha_2} = q_3\,, q_{\alpha_3\alpha_1} = q_2\,, q_{\alpha_2\alpha_3} = q_1\right] =$$

$$= \frac{1}{2} P(q_1)\, x(q_1)\, \delta(q_1 - q_2)\, \delta(q_1 - q_3) + \frac{1}{2} P(q_1)\, P(q_2)\, \Theta(q_1 - q_2)\, \delta(q_2 - q_3) +$$

$$+ \frac{1}{2} P(q_2)\, P(q_3)\, \Theta(q_2 - q_3)\, \delta(q_3 - q_1) + \frac{1}{2} P(q_3)\, P(q_1)\, \theta(q_3 - q_1)\, \delta(q_1 - q_2) +$$

$$\boxed{\begin{array}{l} q_1 = q_2 \leq q_3 \\ q_1 = q_3 \leq q_2 \\ q_2 = q_3 \leq q_1 \end{array}}$$